

In the following exercises, all coordinates and components are given with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$. For the 2-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and for the 3-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

1. (1 point) Select all that apply. Which of the following equalities hold true?

A. $\cos \angle(\mathbf{a}, \mathbf{b}) = \frac{\text{pr}_{\mathbf{a}}^{\perp}(\mathbf{b})}{|\mathbf{b}|}$ B. $\text{Pr}_{\mathbf{a}}^{\perp}(\mathbf{b}) = \text{pr}_{\mathbf{a}}^{\perp}(\mathbf{b}) \frac{\mathbf{a}}{|\mathbf{a}|}$ C. $\sin \angle_{\text{or}}(\mathbf{a}, \mathbf{b}) = \frac{\text{pr}_{\mathbf{j}(\mathbf{a})}^{\perp}(\mathbf{b})}{|\mathbf{b}|}$ D. $\text{Pr}_{\mathbf{b}}^{\perp}(\mathbf{a}) = \frac{\langle \mathbf{b}, \mathbf{a} \rangle}{\langle \mathbf{b}, \mathbf{b} \rangle} \mathbf{b}$

2. (1 point) The values k for which the distance from the point $(2, 3)$ to the line $8x + 15y + k = 0$ equals 5 are _____.

Setup 1: Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

3. (1 point) Consider Setup 1 above. Let P be the orthogonal projection of C on AB . The coordinates of P are _____.
4. (1 point) Consider Setup 1 above. Let $\mathbf{v}$ be the orthogonal projection of $\overrightarrow{CB}$ on $\overrightarrow{CA}$. The components of $\mathbf{v}$ are _____.
5. (1 point) Select all that apply. The scalar product is
A. bilinear B. skew-symmetric C. positive definite D. symmetric
6. (1 point) The coordinates of a point on the line $5x - 4y - 4 = 0$ which lies at equal distance from the points $A(1, 0)$ and $B(-2, 1)$ are _____.
7. (1 point) Consider the vectors $\mathbf{q} = 3\mathbf{i} + \mathbf{j}$ and $\mathbf{p} = \mathbf{i} + 2\mathbf{j} + \lambda\mathbf{k}$ with $\lambda \in \mathbb{R}$. If the cosine of the angle $\angle(\mathbf{p}, \mathbf{q})$ is $5/12$, then $\lambda =$ _____.
8. (1 point) The distance between the planes $x - 2y - 2z + 7 = 0$ and $2x - 4y - 4z + 17 = 0$ is _____.
9. (1 point) The cosine of the angle between the plane $x - \sqrt{2}y + z - 1 = 0$ and the plane $x + \sqrt{2}y - z + 3 = 0$ is _____.
10. (1 point) The coordinates of a point A on the line $\ell : \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{1}$ which lies at distance $\sqrt{3}$ from the plane $x + y + z + 3 = 0$ are _____.
11. (1 point) Consider the vector $\mathbf{v}$ which is perpendicular to $\mathbf{a}(4, -2, -3)$ and to $\mathbf{b}(0, 1, 3)$. If $\mathbf{v}$ describes an acute angle with Ox and $|\mathbf{v}| = 26$ then the components of $\mathbf{v}$ are _____.

Answers:

1. A, B, C, D
2. 24, -146
3. $\left(-\frac{7}{2}, \frac{7}{2}\right)$
4. $\left(-\frac{24}{5}, \frac{12}{5}\right)$
5. A, C, D
6. $\left(-\frac{12}{7}, -\frac{22}{7}\right)$
7. $\lambda = \pm\sqrt{9.4}$
8. $\frac{1}{2}$
9. $\frac{1}{2}$
10. (1, 0, -1) or (-1, -3, -2)
11. (6, 24, -8)